

TITLE - DEVELOP A CYBER TRIAGE TOOL TO STREAMLINE DIGITAL FORENSIC INVESTIGATION

A PROJECT REPORT & FORENSIC TRIAGE REPORT

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in partial fulfillment for the award of the degree of

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IN

Department of Computer Science and Engineering (IoT & Cyber Security including Blockchain)

Sant Dnyaneshwar Shikshan Sanstha's

ANNASAHEB DANGE COLLEGE OF ENGINEERING AND TECHNOLOGY, ASHTA

(An Empowered Autonomous Institute)

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CERTIFICATE

This is to certify that the Report entitled "**DEVELOP A CYBER TRIAGE TOOL TO STREAMLINE DIGITAL FORENSIC INVESTIGATION**" submitted by **PRAGATI KIRAN PATIL (23101024)**, **TANISHA PRAKASH LOHAR (23101009)**, and **GAURAV SUBHASH RAJGURU (23101004)** to Annasaheb Dange College of Engineering and Technology (ADCET), Ashta, Maharashtra, is record of Bonafide Project work carried out by them under my supervision and guidance and is worthy of consideration for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** (IoT, Cyber Security including Blockchain Technology) of the Institute.

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ABSTRACT

The rapid growth of cybercrime and sophisticated cyber attacks has significantly increased the need for efficient digital forensic investigations. Modern organizations generate vast amounts of digital evidence, including disk images, system logs, registry entries, memory dumps, and network traffic, making manual investigation a time-consuming and complex task. Traditional digital forensic tools often require investigators to switch between multiple applications and manually analyze large volumes of evidence, which can delay incident response and increase the possibility of overlooking critical artifacts.

The project, **"Develop a Cyber Triage Tool to Streamline Digital Forensic Investigation,"** designs and implements an intelligent digital forensic and incident response platform that automates evidence collection, multi-parser triage, and reporting. The system supports the acquisition and examination of forensic disk images, system logs, registry data, memory dumps, and network activities to identify suspicious behavior and Indicators of Compromise (IOCs) associated with cyber attacks. Artificial Intelligence (AI) and Machine Learning (ML) techniques are integrated to perform anomaly detection (Isolation Forest), classify suspicious activities (Random Forest), and assign calibrated risk scores to digital artifacts.

The tool features a responsive web interface with interactive dashboards, graphical summaries, investigation timelines, and comprehensive reporting capabilities. Investigation reports are exported in PDF, JSON, STIX 2.1, and CSV formats, allowing seamless sharing with Security Operation Centers (SOCs), law enforcement agencies, and court proceedings. Cryptographic hash chaining is implemented to preserve the chain of custody and guarantee tamper-proof evidence integrity.

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1. EVIDENCE IDENTIFICATION & CRYPTOGRAPHIC INTEGRITY

Target Artifact Filename	ransomware_incident_syslog.log
Acquisition File Size	2,646 bytes
Identified Artifact Type	LOG
SHA-256 Hash (Integrity)	dd0be9cf2a87b54bed5cfee78b9f9dede56aa0b2419c69116f69fb1052a2f57d
MD5 Checksum	615a5d1766f6695f2b4947584cf2a4f0
SHA-1 Checksum	047f58cede63b2291d578b47e79d08eb51de5452
Acquisition Timestamp	2026-08-18 00:02:14
Chain of Custody Integrity	VERIFIED / CRYPTOGRAPHICALLY INTACT

Forensic Preservation Protocol:

Digital evidence was acquired and cryptographically stamped upon ingestion. Bit-level SHA-256, MD5, and SHA-1 hashes were computed prior to analysis. An immutable hash chain links this artifact to the central forensic database, preventing post-acquisition spoliation and satisfying legal admissibility criteria under the Indian Evidence Act (Section 65B).

2. EXECUTIVE SUMMARY & AI THREAT CLASSIFICATION

Overall Threat Risk Score	100/100 (CRITICAL)
Primary Threat Vector (Supervised ML)	Ransomware / Extortion (44.0% Confidence)
Cyber Kill Chain Lifecycle Status	Critical Stage - Active Threat Impact / Ransomware Execution (100.0%)
Unsupervised Anomaly Model (Isolation Forest)	Status: Anomalous (Anomaly Index: 0.00)
Forensic Indicators Quantified	6 Indicators of Compromise 11 High-Risk Triggers

Investigative Summary Narrative:

Automated digital forensic triage of 'ransomware_incident_syslog.log' identified a CRITICAL risk rating (Score: 100/100). The primary threat vector is classified as 'Ransomware / Extortion' (44.0% confidence). Cyber Kill Chain stage reached: 'Critical Stage - Active Threat Impact / Ransomware Execution' (100.0%). Identified 6 IOCs, 3 signature hits, and 5 MITRE techniques. AI Isolation Forest Anomaly marked status as 'Anomalous'.

3. MITRE ATT&CK; MATRIX & KILL CHAIN PROGRESSION

Tactic	Technique ID	Technique Name	Trigger / Evidence
Impact	T1490	Inhibit System Recovery	vssadmin
Impact	T1486	Data Encrypted for Impact	lockbit
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	powershell
Initial Access	T1190	Exploit Public-Facing Application	rce
Defense Evasion	T1070.001	Indicator Removal: Clear Windows Event Logs	wevtutil

Cyber Kill Chain Lifecycle Stages:

- **1. Initial Access & Delivery:** ■ [ACTIVE] - Adversary achieves initial foothold through phishing, exploit public application, or external authentication.
- **2. Execution & Staging:** ■ [ACTIVE] - Execution of malicious scripts, obfuscated PowerShell, or secondary droppers.
- **3. Persistence & Privilege Escalation:** ■ [INACTIVE] - Establishing persistent autostart registry keys, scheduled tasks, or acquiring administrator tokens.
- **4. Credential Access & Discovery:** ■ [INACTIVE] - Dumping memory credentials, LSASS tampering, SAM extraction, or network enumeration.
- **5. Command & Control (C2):** ■ [INACTIVE] - Maintaining outbound communication channels, beaconing, or DNS tunnels to remote servers.
- **6. Impact & Extortion:** ■ [ACTIVE] - System inhibition, volume shadow copy deletion, mass file encryption, and ransom extortion.

4. INDICATORS OF COMPROMISE (IOCs) & THREAT INTEL

Category	Extracted Indicator (Defanged)	Threat Intel & Reputation
IP Address	194[.]26[.]29[.]112	Active Ransomware C2 Gateway (LockBit Affiliate)
IP Address	185[.]220[.]101[.]5	Malicious C2 / TOR Exit Node (LockBit 3.0 / APT29)
URL	hxxp://185[.]220[.]101[.]5/payload/stage2[.]e	HTTP
SHA-256	e3b0c44298fc1c149afb4c8996fb924	Extracted Hash
MD5	7b2c9a6234b071e62a8291a13426e95c	Extracted Hash
LOLBAS Command	powershell.exe -nop -w hidden -enc	High
LOLBAS Command	vssadmin.exe delete shadows	Critical
LOLBAS Command	wmic shadowcopy delete	High
LOLBAS Command	wbadmin.exe delete catalog	High

5. CHRONOLOGICAL ATTACK TIMELINE

Timestamp	Category	Severity	Forensic Event Description
2026-08-17 02:14:22	System Event	Info	Aug 17 02:14:22 DC01-SRV Microsoft-Windows-Security-Aud
2026-08-17 02:14:45	Suspicious Execution / LOLBAS	High	Aug 17 02:14:45 DC01-SRV Microsoft-Windows-PowerShell[4
2026-08-17 02:15:10	Ransomware / Critical Threat	Critical	Aug 17 02:15:10 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:15:33	Ransomware / Critical Threat	Critical	Aug 17 02:15:33 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:16:01	Ransomware / Critical Threat	Critical	Aug 17 02:16:01 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:16:08	System Event	Info	Aug 17 02:16:08 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:16:15	System Event	Info	Aug 17 02:16:15 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:16:22	System Event	Info	Aug 17 02:16:22 DC01-SRV Microsoft-Windows-Sysmon[1]: P
2026-08-17 02:17:40	Ransomware / Critical Threat	Critical	Aug 17 02:17:40 DC01-SRV Microsoft-Windows-Sysmon[11]:
2026-08-17 02:18:10	Ransomware / Critical Threat	Critical	Aug 17 02:18:10 DC01-SRV Microsoft-Windows-Sysmon[3]: N

6. AI DECISION SUPPORT & MITIGATION PLAYBOOK

Prioritized Incident Containment Directives:

- **[Network Containment] (High Urgency):** Block 2 malicious external IP(s) on perimeter firewall and EDR.
- **[Endpoint Isolation] (Immediate Urgency):** Immediately disconnect host from corporate LAN and Wi-Fi to prevent lateral encryption spreading.
- **[Backup Verification] (Immediate Urgency):** Check offline immutable backups and verify integrity of VSS shadow copies on unaffected systems.
- **[Persistence Removal] (Medium Urgency):** Audit and remove unauthorized Windows Registry Run keys, Scheduled Tasks, and anomalous startup services.

Digital Forensic Investigation Checklist:

- ☐ Inspect Master File Table (\$MFT) and USN Journal for mass file renaming activity.
- ☐ Identify initial access vector (Phishing email, RDP brute force, or unpatched VPN/firewall).
- ☐ Inspect AutoRuns / WMI event subscriptions for disguised binaries.
- ☐ Preserve full volatile memory dump (.raw/.dmp) using WinPmem or FTK Imager prior to power down.
- ☐ Generate and record SHA-256 cryptographic hashes for all acquired disk artifacts.
- ☐ Submit extracted malicious file hashes to internal Threat Intelligence (MISP / VirusTotal).

7. TECHNOLOGY STACK & SYSTEM ARCHITECTURE

The Cyber Triage Tool is engineered using modern web technologies, scalable digital forensic engines, and Machine Learning frameworks. The selected stack guarantees automated ingestion, high-speed triage, tamper-proof evidence integrity, and standardized reporting.

Component Layer	Technologies Utilized
Frontend	HTML5, CSS3 Tactical Dark SOC UI, JavaScript (ES6+), Chart.js 4.4, HTML5 Physics Correlation Graph Canvas
Backend Server	Python 3.11+, Flask Web Framework, RESTful API v1, Gunicorn WSGI, Flask-CORS
Database & Integrity	SQLite 3 with Blockchain-inspired Cryptographic Merkle/Hash Chaining for Chain of Custody
AI & Machine Learning	Scikit-learn (Isolation Forest for Unsupervised Anomaly Detection, Random Forest for Threat Classification)
Digital Forensics	Heuristic YARA Signature Engine, Shannon Binary Entropy Calculator, PE Header Analyzer, EVTX/Sysmon Parser, PCAP Flow Parser, Registry Hive Analyzer
Threat Intelligence	Offline IOC Reputation Feed (LockBit 3.0, APT29, Cobalt Strike, TOR Exit Nodes, CVSS 3.1 Catalog)
Reporting Engine	ReportLab PDF Generator (ADCET Institutional Format), STIX 2.1 Threat Intel Exporter, JSON, CSV
Deployment	Multi-Device Network Access (0.0.0.0), Cloudflare Public Tunnel, Render.com 24/7 Cloud Support

8. PROJECT TIMELINE & MILESTONES

Phase	Activity	Duration	Deliverables
Phase 1	Literature Survey & Requirement Analysis	Week 1–2	Problem identification, requirement specification
Phase 2	System Design & Architecture	Week 3–4	System architecture, DFD, UML diagrams
Phase 3	UI/UX Design	Week 5–6	Dashboard design, wireframes, frontend prototype
Phase 4	Backend Development	Week 7–10	Database schema, RESTful APIs, Flask backend
Phase 5	Evidence Collection & Hash Engine	Week 11–13	Evidence upload, SHA-256 hashing, chain of custody
Phase 6	AI & Forensic Analysis Modules	Week 14–17	Isolation Forest, Random Forest, YARA, MITRE mapper
Phase 7	Dashboard & Report Generation	Week 18–19	Interactive SOC UI, ReportLab institutional PDF
Phase 8	Testing & Validation	Week 20–21	End-to-end testing, bug fixing, test suite
Phase 9	Documentation & Deployment	Week 22–24	Project synopsis, PPT, cloud deployment

9. APPLICATIONS & SUSTAINABLE DEVELOPMENT GOALS

- 1. Law Enforcement Agencies:** Investigates cybercrimes such as ransomware attacks, digital fraud, and unauthorized access while maintaining forensic evidence integrity.
- 2. Security Operation Centers (SOCs):** Accelerates tier-1 and tier-2 incident response by automatically triaging logs, network flows, and endpoints in real time.
- 3. Government & Defense Organizations:** Protects critical digital infrastructure, detects Advanced Persistent Threats (APTs), and secures confidential networks.
- 4. Financial & Banking Institutions:** Rapidly identifies online banking fraud, unauthorized transactions, malware injections, and insider data theft.
- 5. Healthcare Organizations:** Investigates attacks targeting Electronic Health Records (EHR) and ensures patient data confidentiality and uninterrupted care.

United Nations Sustainable Development Goals (SDG) Mapping:

SDG Goal	Project Contribution
SDG 9: Industry, Innovation and Infrastructure	Promotes AI-driven digital forensic innovation, automated cyber triage, and strengthens national cybersecurity infrastructure.
SDG 16: Peace, Justice and Strong Institutions	Supports lawful cybercrime investigation, digital evidence integrity, rule of law, and transparent digital forensic accountability.

10. REFERENCES

- [1] S. Li and Y. Liu, "Human-Centric Artificial Intelligence Enabled Digital Images and Videos Forensic Triage," in *Proc. IEEE International Conference on Human-Centered Cognitive Systems (HCCS)*, Cardiff, U.K., Dec. 2023, pp. 1–5, doi: 10.1109/HCCS59561.2023.10452651.
- [2] D. Dunsin, M. C. Ghanem, K. Ouazzane, and V. Vassilev, "A Comprehensive Analysis of the Role of Artificial Intelligence and Machine Learning in Modern Digital Forensics and Incident Response," *Forensic Science International: Digital Investigation*, vol. 48, Art. no. 301675, 2024, doi: 10.1016/j.fsidi.2023.301675. (Scopus Indexed)
- [3] S. Nath, K. Summers, J. Baek, and G. J. Ahn, "Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics," in *Proc. IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA)*, 2024, pp. 11–20, doi: 10.1109/TPS-ISA62245.2024.00012. (IEEE & Scopus Indexed)
- [4] Y. Zhang, X. Fu, B. Luo, and X. Du, "A Crowdsourcing Digital Forensics Platform for IoT Environments Powered by Blockchain," in *Proc. IEEE International Conference on Communications (ICC)*, Denver, CO, USA, 2024, pp. 2016–2021, doi: 10.1109/ICC51166.2024.10622399. (IEEE & Scopus Indexed)
- [5] M. Ivanova and S. Stefanov, "Regarding Artificial Intelligence in Digital Forensic Investigation: Applications and Solutions," in *Proc. XXXIII International Scientific Conference Electronics (ET)*, 2024, pp. 1–6, doi: 10.1109/ET63133.2024.10721531.
- [6] A. Bhardwaj and S. K. Choudhary, "AI Based Decision Support System for Cyber Forensics Investigations," in *Proc. IEEE 4th International Conference on ICT in Business, Industry & Government (ICTBIG)*, 2024, doi: 10.1109/ICTBIG64922.2024.10911602.
- [7] M. Z. H. Zamil and T. M. Khan, "AI-Driven Digital Evidence Triage in Digital Forensics: A Comprehensive Review," in *Proc. 13th International Symposium on Digital Forensics and Security (ISDFS)*, Boston, MA, USA, 2025, doi: 10.1109/ISDFS65363.2025.11011908.
- [8] M. A. Alshurman et al., "Enhancing Digital Investigation: The Role of Generative AI in Evidence Identification and Analysis," in *Proc. 12th International Conference on Information Technology (ICIT)*, Amman, Jordan, 2025, doi: 10.1109/ICIT64950.2025.11049120.

11. CONCLUSION & DECLARATION

The proposed Cyber Triage Tool addresses the critical challenges of modern digital forensic investigations by providing a unified, automated, and AI-assisted triage platform. By integrating multi-source evidence acquisition, YARA heuristic scanning, Shannon entropy analysis, supervised Machine Learning classification (Random Forest), unsupervised anomaly detection (Isolation Forest), MITRE ATT&CK; correlation, and blockchain-inspired cryptographic hash chaining, the tool significantly reduces incident triage latency while ensuring complete evidentiary integrity. The automated multi-format reporting engine standardizes forensic outputs, bridging the gap between technical analysts and decision-makers.

To the best of our knowledge, the above-submitted information is true and correct.

Date: 18-09-2026

Place: ADCET, Ashta

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